Md. Mehadi Hassan

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Professional Summary

Hands-on AI Engineer specializing in designing and deploying production-style AI applications powered by Large Language Models (LLMs) and advanced Retrieval-Augmented Generation (RAG). Highly skilled in complex multi-agent orchestration using LangGraph, system tool integrations, and comprehensive harness engineering. Proficient in integrating cutting-edge LLM APIs and building high-performance, vector-search-backed conversational systems using Python, Pinecone, and FAISS. Focused on shipping scalable, robust, and industry-grade AI automation workflows in distributed remote environments.

Technical Skills

- **Generative AI & Multi-Agent Orchestration:** LangGraph, LangChain, Multi-Agent Systems, Harness Engineering, Tool Systems Integration, Prompt Engineering
- **Vector Search & RAG Infrastructure:** Chroma, Pinecone, FAISS, Advanced Retrieval Pipelines, Context Chunking
- **LLM Providers & APIs:** Groq, Google Generative AI (Gemini), REST APIs Integration
- **Frameworks & Deployment Tools:** Streamlit, Flask, Docker, Git, Linux/Bash
- **ML/NLP Foundations & Core Languages:** Python, Scikit-learn, Hugging Face Transformers, SQL, Bash
- **Data Engineering & Databases:** Pandas, NumPy, Matplotlib, Seaborn, MySQL, MongoDB

Projects

MehadiAI – Retrieval-Augmented Agent Toolkit | GitHub

- Architected a modular RAG-based agent toolkit for building LLM-powered applications with tool/plugin integrations and semantic document search.
- Built Python backend services and a responsive front-end to deliver interactive, LLM-driven assistants and automation workflows.
- Designed reusable components for connector integrations, retrieval pipelines, and multi-step agent orchestration.

Multi-Agent Research System | GitHub

- Engineered a multi-agent AI system coordinating several LLM agents to perform automated web search, scraping, and structured report generation.
- Implemented agent orchestration and inter-agent workflows with LangChain, enabling live retrieval and automated critique between agents.
- Built a Streamlit interface exposing the multi-agent pipeline as a usable, end-to-end AI application.

Medical Chatbot | GitHub

- Built a production-style RAG chatbot combining LangChain orchestration, Pinecone vector search, and Groq-hosted LLMs for domain-specific Q&A.
- Integrated Hugging Face embeddings to ground LLM responses in retrieved medical documents, reducing hallucination risk.
- Implemented the full AI application stack – data ingestion, indexing, retrieval, and conversational interface – end to end.

Personal Voice Assistant System | GitHub

- Built an AI-powered voice assistant integrating speech recognition, text-to-speech, and Google Generative AI for natural conversational interaction.
- Engineered command-parsing and action-execution modules enabling web automation, Wikipedia search, music playback, and task execution.
- Focused on reliable audio I/O and an extendable architecture for adding new AI-driven capabilities.

Education

IUBAT – International University of Business Agriculture and Technology
Bachelor of Science in Computer Science and Engineering (CSE)

Certifications

- **AI Agent Development Bootcamp for Programmers** – Ostad | View Certificate
- **Full Stack Data Science with Generative AI** – InceptionBD | View Certificate